Supplementary Materials for "Persistent Dealer-Gamma Regimes Under Temporal Obfuscation: A Re-Run on Complete Inputs with a Controlled LLM Re-Test"

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S1 Series Definitions in SQL

The daily series of Table 1 of the article are defined by the four read-only statements of Sections S1.1 to S1.4, which were run on the project database on 2026-10-28; Section S1.5 defines the derived and recovered series. In the statements `:sym` is the symbol (SPY, QQQ or IWM) and `:y0`, `:y1` bound one calendar year, because the raw table is read one symbol-year per statement. Every statement stops at 2026-10-09, the last day before the vendor switch (Section 3.3 of the article). The first two statements return the 30 daily values of every window. The pipeline stores the daily rows and cuts the windows in a second step (Section 3.4 of the article); unlike that step, the first statement drops a window when one of its 30 days of the `gex.options_daily_summary` calendar is missing from the archive table.

S1.1 Pre-Rebuild Archive Table (`old_bs`)

```
-- old_bs: pre-rebuild archive table (read-only). Parameter :sym is the symbol.
-- Calendar: gex.options_daily_summary for the symbol; a 30-day window is valid only if
↪ all 30 consecutive calendar days exist in the archive table.
-- The archive ends on 2026-10-02, so the last window ends there; IWM and QQQ have no
↪ rows from 2020-12-03 to 2025-12-30.
WITH cal AS (SELECT trading_date, row_number() OVER (ORDER BY trading_date) AS rn
             FROM gex.options_daily_summary WHERE symbol = :'sym' AND trading_date <=
↪ DATE '2026-10-09'),
     f AS (SELECT c.rn, c.trading_date, z.total_gex, z.net_call_gex, z.net_put_gex, z.
↪ underlying_price
           FROM cal c JOIN backups.options_daily_summary_pre_rebuild_20261004 z
             ON z.symbol = :'sym' AND z.trading_date = c.trading_date)
SELECT e.trading_date AS window_end,
       array_agg(w.total_gex     ORDER BY w.rn) AS total_gex_30,
       array_agg(w.net_call_gex  ORDER BY w.rn) AS call_gex_30,
       array_agg(w.net_put_gex   ORDER BY w.rn) AS put_gex_30
FROM f e JOIN f w ON w.rn BETWEEN e.rn - 29 AND e.rn
GROUP BY e.trading_date, e.rn
HAVING count(*) = 30          -- hole-free window
ORDER BY e.rn;
```

S1.2 Rebuilt Daily Table (`new_full`)

```
-- new_full: stored per-contract gamma, same-day expiries included (gex.
↪ options_daily_summary, rebuilt 2026-10-08). Parameter :sym is the symbol.
WITH s AS (SELECT trading_date, total_gex, net_call_gex, net_put_gex, underlying_price,
                  row_number() OVER (ORDER BY trading_date) AS rn
           FROM gex.options_daily_summary
           WHERE symbol = :'sym' AND trading_date <= DATE '2026-10-09')      --
↪ alpha_vantage ends 2026-10-09; cboe_delayed starts 2026-10-12
SELECT e.trading_date AS window_end,
       array_agg(w.total_gex    ORDER BY w.rn) AS total_gex_30,
```

```
       array_agg(w.net_call_gex ORDER BY w.rn) AS call_gex_30,
       array_agg(w.net_put_gex  ORDER BY w.rn) AS put_gex_30
FROM s e JOIN s w ON w.rn BETWEEN e.rn - 29 AND e.rn
WHERE e.rn >= 30
GROUP BY e.trading_date, e.rn
HAVING count(*) = 30
ORDER BY e.rn;
```

S1.3 Raw Chains Without Same-Day Expiries (`new_ex0dte`)

```
-- new_ex0dte: raw chains, one symbol-year per statement. Parameters :sym, :y0, :y1 (SPY
↪  and IWM 2008 to 2026, QQQ 2011 to 2026).
-- enable_seqscan=off is a session planner setting (no writes): the index plan is faster
↪  and returns the same numbers (largest relative difference 5e-13).
SET enable_seqscan = off;
SELECT trading_date,
       count(*) FILTER (WHERE expiration > trading_date)        AS n_contracts,
       count(*) FILTER (WHERE expiration = trading_date)        AS n_0dte,
       sum(CASE WHEN option_type = 'call' THEN g END)  FILTER (WHERE expiration >
↪ trading_date) AS call_gex_ex0dte,
      -sum(CASE WHEN option_type = 'put'  THEN g END)  FILTER (WHERE expiration >
↪ trading_date) AS put_gex_ex0dte,
       sum(CASE WHEN option_type = 'call' THEN g ELSE -g END) FILTER (WHERE expiration >
↪  trading_date) AS total_gex_ex0dte,
       sum(CASE WHEN option_type = 'call' THEN g ELSE -g END) FILTER (WHERE expiration =
↪  trading_date) AS total_gex_0dte,
       sum(CASE WHEN option_type = 'call' THEN g ELSE -g END)
↪                AS total_gex_all  -- ties to gex.options_daily_summary.total_gex (rel.
↪  diff. < 3e-7)
FROM (SELECT trading_date, option_type, expiration,
             gamma::float8 * open_interest::float8 * 100.0 * underlying_price::float8 *
↪ underlying_price::float8 * 0.01 AS g   -- float8: real*int would stay float4
      FROM gex.options_chains_partitioned
      WHERE symbol = :'sym' AND trading_date >= :'y0' AND trading_date < :'y1'
        AND trading_date <= DATE '2026-10-09'                      -- vendor cap:
↪ alpha_vantage ends 2026-10-09, cboe_delayed starts 2026-10-12
        AND gamma IS NOT NULL AND open_interest > 0 AND underlying_price > 0) t
GROUP BY trading_date
ORDER BY trading_date;
```

Since the database additions of 2026-11-03 (Section 3.3 of the article) the pipeline's statement also takes the spot from `gex.spot_fill_log` on the days listed there, whose chain rows have no spot: it reads `coalesce(c.underlying_price, f.spot_used)` from a left join of the chains with that table on symbol and trading date (rows with `in_summary`), and keeps the row filter otherwise. On every other day the result is unchanged.

S1.4 Late-Session Intraday Snapshot (`intraday_late`)

The statement as listed returns Variant 1 of its comments, the `market_close` slot, which equals the end-of-day row. The series of the study is Variant 2: the filter on the clock time is replaced by a filter on the snapshot type (`snapshot_type` equal to `final_rush`, 19:55:59 UTC), so that the snapshot is selected by type and not by clock time, and the three irregular days 2026-07-29, 2026-08-03 and 2026-09-14 are dropped when the series is built.

```
-- intraday_late: late-session snapshot, SPY, 2026-07-27 to 2026-10-09 (EDT: 20:00 UTC =
↪  16:00 ET).
-- Variant 1 'late' = last snapshot with timestamp < 20:01:00 UTC  -> the market_close
↪ slot (20:00:59), which EQUALS the daily summary row (not independent).
-- Variant 2, used in this study (independent of the end-of-day row): add  AND
↪ snapshot_type = 'final_rush'  (19:55:59 UTC).
SELECT DISTINCT ON (d)
       d AS trading_date, snapshot_timestamp, snapshot_type, total_gex, net_call_gex,
↪ net_put_gex, underlying_price, total_contracts
FROM (SELECT *, (snapshot_timestamp AT TIME ZONE 'America/New_York')::date AS d
      FROM gex.intraday_gex_summary
      WHERE symbol = 'SPY'
```

```
        AND snapshot_timestamp >= TIMESTAMPTZ '2026-07-27 00:00+00' AND
↪ snapshot_timestamp < TIMESTAMPTZ '2026-10-10 00:00+00'
        AND snapshot_type <> 'test'
        AND (snapshot_timestamp AT TIME ZONE 'UTC')::time < TIME '20:01:00') x
ORDER BY d, snapshot_timestamp DESC;
```

S1.5 Derived and Recovered Series

Call leg only (`call_only`). The sum of the call terms `g` of the third statement over all expiries, same-day expiries included, with the put terms dropped. In the rebuilt summary table this is `net_call_gex` (up to the float4 rounding of the stored column), which the second statement returns as `call_gex_30`.

Scaled series (`new_full_scaled`). The `new_full` values of QQQ and IWM multiplied by one fixed factor per symbol (Section 3.6 of the article), which the following statement computes (tested read-only on 2026-10-29). The factor and the number of days are stored with the series.

```
-- new_full_scaled: factor of one symbol = median over 2020-01-01..2026-10-09 of the
↪ ratio of SPY's open-interest notional to the symbol's.
-- Parameter :sym is QQQ or IWM; the scaled values are the new_full values of the symbol
↪  times this factor.
WITH n AS (SELECT symbol, trading_date, total_weighted_oi::float8 * 100.0 *
↪ underlying_price::float8 AS notional
          FROM gex.options_daily_summary
          WHERE symbol IN ('SPY', :'sym') AND trading_date BETWEEN DATE '2020-01-01'
↪ AND DATE '2026-10-09')
SELECT percentile_cont(0.5) WITHIN GROUP (ORDER BY s.notional / q.notional) AS factor,
       count(*) AS n_days
FROM n s JOIN n q ON q.trading_date = s.trading_date AND s.symbol = 'SPY' AND q.symbol =
↪  :'sym'
WHERE q.notional > 0;
```

Recovered input series. The series `jan2026_phase4a`, `p3_2024` and `p4_2020` (and the dated twins of the last two) are not defined by SQL. They are the daily values parsed from the data lines of the surviving batch input files `phase4a_postgresql_20260104_184832.jsonl`, `phase3_full_2024_baseline_v2.jsonl` and `batch_regime_20251120_104630.jsonl` (folder `reports/validation/paper2_regime_windows/batch_jobs/`); overlapping windows are checked for consistent values (Section 4.1 of the article).

S2 Run Ledger

Tables S1 and S2 list every Batch job and the backfilled runs of the study: the identifiers, the token usage and the cost. Tables S3, S4 and S5 list the local runs of the open-weights models gpt-oss-20b, Llama 3.1 8B and Qwen3-8B (Section 3.5 of the article), which cost nothing. The arms are defined in Section 3.6 of the article.

Table S1: Run ledger: one row per Batch job, with the token usage and the cost priced at Batch rates. The identifier has the form `<arm>_<model>_<mode>_<UTC timestamp>_c<chunk>`. Only runs paid in this study enter the total; the synchronous pilot (`sync`) is included in it. The eight jobs with the timestamp 20261103 are the re-requests of windows whose input changed after the database additions of 2026-11-03 (Section 3.7 of the article): 1,049 of their 1,121 answers replace answers of the earlier jobs, which stay in the ledger and in the total. The three backfilled o4-mini runs are listed below it, as stored answers of earlier runs that are not spend of this study.

Run identifier	Requests	Prompt tokens	Completion tokens	Cost (USD)	Status
bridge_jan_gpt-6-luna_batch_20261029T203742Z_c0	1,476	2,823,588	1,086,712	0.41	ingested
bridge_jan_gpt-6-luna_batch_20261029T203742Z_c1	1,476	2,823,588	1,082,280	0.41	ingested
full_spy_gpt-6-luna_batch_20261029T203745Z_c0	1,227	2,347,251	871,277	0.34	ingested
full_spy_gpt-6-luna_batch_20261029T203745Z_c1	1,226	2,345,338	969,842	0.36	ingested
full_spy_gpt-6-luna_batch_20261029T203745Z_c2	1,226	2,345,338	1,004,850	0.37	ingested
full_spy_gpt-6-luna_batch_20261029T203745Z_c3	1,226	2,345,338	1,024,540	0.37	ingested
full_spy_gpt-6-luna_batch_20261103T040042Z_c0	348	665,724	272,121	0.10	ingested
ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c0	1,227	2,347,251	839,753	0.33	ingested
ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c1	1,226	2,345,338	968,965	0.36	ingested
ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c2	1,226	2,345,338	951,505	0.36	ingested
ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c3	1,226	2,345,338	982,560	0.36	ingested
ex0dte_spy_gpt-6-luna_batch_20261103T040045Z_c0	332	635,116	263,091	0.10	ingested
p34_obf_2024_gpt-6-luna_batch_20261029T203752Z_c0	234	447,642	156,636	0.06	ingested
p34_dated_2024_gpt-6-luna_batch_20261029T203754Z_c0	234	461,682	154,330	0.06	ingested
p34_obf_2020_gpt-6-luna_batch_20261029T203757Z_c0	234	447,642	132,619	0.06	ingested
p34_dated_2020_gpt-6-luna_batch_20261029T203759Z_c0	234	461,682	130,805	0.06	ingested
full_qqq_gpt-6-luna_batch_20261029T204029Z_c0	1,354	2,590,202	1,012,074	0.38	ingested
full_qqq_gpt-6-luna_batch_20261029T204029Z_c1	1,353	2,588,289	1,004,705	0.38	ingested
full_qqq_gpt-6-luna_batch_20261029T204029Z_c2	1,353	2,588,289	1,057,334	0.39	ingested
full_qqq_gpt-6-luna_batch_20261103T040049Z_c0	134	256,342	107,963	0.04	ingested
full_iwm_gpt-6-luna_batch_20261029T204033Z_c0	1,228	2,349,164	762,229	0.31	ingested

Table S2: Run ledger (continued): remaining Batch jobs, the total of this study and the backfilled runs. Rows are rounded to cents; the totals are computed from the unrounded values (the three backfilled runs sum to 9.97).

Run identifier	Requests	Prompt tokens	Completion tokens	Cost (USD)	Status
full_iwm_gpt-6-luna_batch_20261029T204033Z_c1	1,228	2,349,164	719,759	0.30	ingested
full_iwm_gpt-6-luna_batch_20261029T204033Z_c2	1,228	2,349,164	826,909	0.32	ingested
full_iwm_gpt-6-luna_batch_20261029T204033Z_c3	1,228	2,349,164	855,981	0.33	ingested
full_iwm_gpt-6-luna_batch_20261103T040052Z_c0	184	351,992	113,134	0.05	ingested
full_qqq_scaled_gpt-6-luna_batch_20261029T204036Z_c1	877	1,677,701	750,362	0.27	ingested
full_qqq_scaled_gpt-6-luna_batch_20261029T204036Z_c0	877	1,677,701	694,967	0.26	ingested
full_qqq_scaled_gpt-6-luna_batch_20261103T040056Z_c0	31	59,303	28,589	0.01	ingested
full_iwm_scaled_gpt-6-luna_batch_20261029T204039Z_c0	878	1,679,614	614,557	0.24	ingested
full_iwm_scaled_gpt-6-luna_batch_20261029T204039Z_c1	877	1,677,701	695,367	0.26	ingested
full_iwm_scaled_gpt-6-luna_batch_20261103T040059Z_c0	30	57,390	26,336	0.01	ingested
ex0dte_qqq_gpt-6-luna_batch_20261030T010447Z_c1	892	1,706,396	680,360	0.26	ingested
ex0dte_qqq_gpt-6-luna_batch_20261030T010447Z_c0	892	1,706,396	622,593	0.24	ingested
ex0dte_qqq_gpt-6-luna_batch_20261103T040102Z_c0	30	57,390	20,013	0.01	ingested
ex0dte_iwm_gpt-6-luna_batch_20261030T010451Z_c0	893	1,708,309	560,829	0.23	ingested
ex0dte_iwm_gpt-6-luna_batch_20261030T010451Z_c1	892	1,706,396	616,040	0.24	ingested
ex0dte_iwm_gpt-6-luna_batch_20261103T040105Z_c0	32	61,216	20,550	0.01	ingested
intraday_spy_gpt-6-luna_batch_20261029T204042Z_c0	54	103,302	45,525	0.02	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c0	1,218	2,330,034	813,090	0.32	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c1	1,218	2,330,034	970,974	0.36	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c2	1,217	2,328,121	953,927	0.35	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c3	1,217	2,328,121	830,288	0.32	ingested
pilot_bridge_jan_gpt-6-luna_sync_20261029T203454Z_c0	40	76,520	28,015	0.02	ingested
Total (this study)	35,833	68,576,609	26,324,356	10.02	
Backfilled o4-mini runs (not spend of this study)					
r1_o4mini_phase4a_20260104	1,476	2,655,324	2,928,992	7.90	backfill
r1_p3_o4mini_20251120	223	414,557	405,127	1.12	backfill
r1_p4_o4mini_20251120	223	414,557	325,436	0.94	backfill

Table S3: Run ledger (continued): the local runs of gpt-oss-20b, one row per chunk, answered by vLLM 0.32.0 on one NVIDIA A100 GPU each. The GPU minutes include the start of the engine and depend on the load that other users put on the shared GPU (the two longest runs shared their GPU with other jobs). The four runs marked *retry* request again the answers that ended at the limit of 4,000 completion tokens, at 8,000 tokens; their answers replace the cut-off ones. The run with the timestamp 20261107T205228Z holds six test-retest repeats that a second generation step added while other chunks of the arm were still running, so `oss_ex0dte_spy` carries 241 repeats instead of 235.

Run identifier	Requests	Prompt tokens	Completion tokens	GPU minutes	Status
`pilot_oss_bridge_jan_gpt-oss-20b_local_20261107T180914Z_c0`	40	74,400	66,286	4.1	ingested
`oss_bridge_jan_gpt-oss-20b_local_20261107T181402Z_c0`	775	1,441,500	1,657,168	114.8	ingested
`oss_bridge_jan_gpt-oss-20b_local_20261107T181402Z_c1`	775	1,441,500	980,476	28.3	ingested
`oss_bridge_jan_gpt-oss-20b_local_20261107T222800Z_c0`	3	5,580	6,554	2.6	retry
`oss_full_spy_gpt-oss-20b_local_20261107T184402Z_c0`	1,232	2,291,520	2,260,858	46.1	ingested
`oss_full_spy_gpt-oss-20b_local_20261107T184402Z_c1`	1,232	2,291,520	2,501,916	27.7	ingested
`oss_full_spy_gpt-oss-20b_local_20261107T184402Z_c2`	1,231	2,289,660	2,504,596	26.8	ingested
`oss_full_spy_gpt-oss-20b_local_20261107T184402Z_c3`	1,231	2,289,660	2,559,220	27.0	ingested
`oss_full_spy_gpt-oss-20b_local_20261107T222803Z_c0`	7	13,020	14,541	2.7	retry
`oss_ex0dte_spy_gpt-oss-20b_local_20261107T191022Z_c0`	1,232	2,291,520	2,157,801	64.7	ingested
`oss_ex0dte_spy_gpt-oss-20b_local_20261107T191022Z_c1`	1,232	2,291,520	2,401,169	60.4	ingested
`oss_ex0dte_spy_gpt-oss-20b_local_20261107T191022Z_c2`	1,231	2,289,660	2,322,398	122.7	ingested
`oss_ex0dte_spy_gpt-oss-20b_local_20261107T191022Z_c3`	1,231	2,289,660	2,406,970	44.4	ingested
`oss_ex0dte_spy_gpt-oss-20b_local_20261107T205228Z_c0`	6	11,160	8,319	1.8	ingested
`oss_ex0dte_spy_gpt-oss-20b_local_20261107T223708Z_c0`	2	3,720	3,635	1.7	retry
`oss_p34_obf_2024_gpt-oss-20b_local_20261107T181626Z_c0`	234	435,240	325,779	8.7	ingested
`oss_p34_dated_2024_gpt-oss-20b_local_20261107T182922Z_c0`	234	449,280	378,043	5.4	ingested
`oss_p34_dated_2024_gpt-oss-20b_local_20261107T222807Z_c0`	2	3,840	3,684	2.3	retry
`oss_p34_obf_2020_gpt-oss-20b_local_20261107T183458Z_c0`	234	435,240	263,783	4.3	ingested
`oss_p34_dated_2020_gpt-oss-20b_local_20261107T183929Z_c0`	234	449,280	292,327	4.3	ingested
Total (local)	12,398	23,088,480	23,115,523	600.7	

Table S4: Run ledger (continued): the local runs of Llama 3.1 8B, one row per chunk, answered by vLLM 0.32.0 on one NVIDIA A100 GPU each, as in Table S3. No answer reached the token limit, so there are no retry runs.

Run identifier	Requests	Prompt tokens	Completion tokens	GPU minutes	Status
pilot_llama_bridge_jan_ llama-31-8b-instruct_local_ 20261108T174407Z_c0	40	72,800	6,712	3.1	ingested
llama_bridge_jan_ llama-31-8b-instruct_local_ 20261108T175235Z_c0	775	1,410,500	131,970	29.9	ingested
llama_bridge_jan_ llama-31-8b-instruct_local_ 20261108T175235Z_c1	775	1,410,500	129,347	8.4	ingested
llama_full_spy_ llama-31-8b-instruct_local_ 20261108T175252Z_c0	1,232	2,242,240	212,867	17.3	ingested
llama_full_spy_ llama-31-8b-instruct_local_ 20261108T175252Z_c1	1,232	2,242,240	210,077	17.4	ingested
llama_full_spy_ llama-31-8b-instruct_local_ 20261108T175252Z_c2	1,231	2,240,420	208,263	9.4	ingested
llama_full_spy_ llama-31-8b-instruct_local_ 20261108T175252Z_c3	1,231	2,240,420	208,043	7.7	ingested
llama_ex0dte_spy_ llama-31-8b-instruct_local_ 20261108T175256Z_c0	1,232	2,242,240	212,383	28.8	ingested
llama_ex0dte_spy_ llama-31-8b-instruct_local_ 20261108T175256Z_c1	1,232	2,242,240	209,825	17.5	ingested
llama_ex0dte_spy_ llama-31-8b-instruct_local_ 20261108T175256Z_c2	1,231	2,240,420	207,502	18.7	ingested
llama_ex0dte_spy_ llama-31-8b-instruct_local_ 20261108T175256Z_c3	1,231	2,240,420	208,475	18.6	ingested
llama_p34_obf_2024_ llama-31-8b-instruct_local_ 20261108T175239Z_c0	234	425,880	39,334	6.5	ingested
llama_p34_dated_2024_ llama-31-8b-instruct_local_ 20261108T175242Z_c0	234	439,920	39,353	5.0	ingested
llama_p34_obf_2020_ llama-31-8b-instruct_local_ 20261108T175244Z_c0	234	425,880	39,178	5.2	ingested
llama_p34_dated_2020_ llama-31-8b-instruct_local_ 20261108T175248Z_c0	234	439,920	39,430	5.0	ingested
Total (local)	12,378	22,556,040	2,102,759	198.4	

Table S5: Run ledger (continued): the local runs of Qwen3-8B, answered by vLLM 0.32.0 as in Table S3, one row per run with its chunks summed (the first passes ran in chunks of about 150 requests). The runs marked *retry* request again, at 8,000 tokens, the answers that ended at the limit of 4,000 completion tokens (10,022 of the 12,378 first-pass requests); their answers replace the cut-off ones. The GPU minutes depend on the load of the shared GPUs: the engine took 27% to 45% of the memory of a GPU that other users were also using, and the retry runs, whose answers are the longest, ran in a few large chunks.

Run identifier	Chunks	Requests	Prompt tokens	Completion tokens	GPU minutes	Status
`pilot_qwen_bridge_jan_qwen3-8b_local_20261108T180454Z`	1	40	76,798	135,241	22.9	ingested
`pilot_qwen_bridge_jan_qwen3-8b_local_20261108T183430Z`	1	23	43,956	145,048	39.9	retry
`qwen_bridge_jan_qwen3-8b_local_20261108T192156Z`	11	1,550	2,975,715	5,358,907	439.9	ingested
`qwen_bridge_jan_qwen3-8b_local_20261109T024816Z`	1	943	1,802,285	5,767,338	105.9	retry
`qwen_full_spy_qwen3-8b_local_20261108T192201Z`	33	4,926	9,405,419	19,293,423	874.0	ingested
`qwen_full_spy_qwen3-8b_local_20261109T091006Z`	3	4,330	8,267,043	25,928,654	1,194.9	retry
`qwen_ex0dte_spy_qwen3-8b_local_20261108T192208Z`	33	4,926	9,398,225	19,187,223	821.2	ingested
`qwen_ex0dte_spy_qwen3-8b_local_20261109T075954Z`	3	4,246	8,100,071	25,024,896	1,901.6	retry
`qwen_p34_obf_2024_qwen3-8b_local_20261108T192142Z`	2	234	453,258	687,018	64.7	ingested
`qwen_p34_obf_2024_qwen3-8b_local_20261108T203436Z`	1	93	180,141	586,172	39.2	retry
`qwen_p34_dated_2024_qwen3-8b_local_20261108T192145Z`	2	234	490,698	740,581	66.8	ingested
`qwen_p34_dated_2024_qwen3-8b_local_20261109T074945Z`	1	99	207,603	630,281	83.5	retry
`qwen_p34_obf_2020_qwen3-8b_local_20261108T192148Z`	2	234	446,296	836,123	97.4	ingested
`qwen_p34_obf_2020_qwen3-8b_local_20261108T214528Z`	1	115	219,345	496,658	16.3	retry
`qwen_p34_dated_2020_qwen3-8b_local_20261108T192151Z`	2	234	483,731	901,531	119.9	ingested
`qwen_p34_dated_2020_qwen3-8b_local_20261108T231634Z`	1	173	357,629	847,291	19.2	retry
Total (local)	98	22,400	42,908,213	106,566,385	5,907.3	

S3 Open-Weights Models: Detail

Section 4.12 of the article compares the three open-weights models in one table. This section gives the detail per model: the rates of gpt-oss-20b next to the rule and gpt-6-luna (Table S6), the verdicts and the reported metrics of Llama 3.1 8B (Table S7), and the answered windows of Qwen3-8B by the verdict of the rule, with the bounds of its detection rate (Table S8). The runs are listed in Tables S3–S5. Rates and agreements are over the first pass of each arm; the repeats enter only the test-retest counts.

S3.1 gpt-oss-20b

Fourteen first-pass answers of gpt-oss-20b ended at the limit of 4,000 completion tokens; each was requested again at 8,000 tokens, and every retry parsed. The two windows without a parsed answer are windows of `oss_full_spy` whose answer closes the reasoning field with a typographic quotation mark followed by control tokens of the model, so that no parser recovers the JSON object; one answer of `oss_bridge_jan` with the same defect was recovered by the fallback parser. The answers are longer than those of gpt-6-luna: between 1,126 and 1,995 completion tokens per arm on average, against 736 to 788 on the bridge and the complete series. On the published

inputs gpt-oss-20b detects every window that o4-mini detects and adds 6 detections where gpt-6-luna adds 5 (exact McNemar test at stride one $p = 0.031$); its rise from 2020–2023 to 2024–2025 is 82.4 percentage points (quarter-block bootstrap $p = 0.0005$). On the complete series its change between the same periods is −10.1 percentage points (bootstrap $p = 0.103$), against −10.2 for the rule ($p = 0.108$). Without same-day expiries it detects 75 of the 502 windows of 2024 and 2025 (14.9%), against 14.7% for the rule and 15.3% for gpt-6-luna, so for this model, too, removing them does not restore the published rates.

gpt-oss-20b and gpt-6-luna give the same verdict in 99.4% of the windows of the complete series (4,660 of 4,689; kappa 0.973) and in 99.6% without same-day expiries (4,670 of 4,691; kappa 0.976). Where they differ, the rule sides with gpt-oss-20b in 16 and with gpt-6-luna in 13 of the 29 windows of `new_full`, and in 14 and 7 of the 21 windows of `new_ex0dte`. On the windows that both answered, the two models detect almost the same number (gpt-6-luna 616 and gpt-oss-20b 617 on `new_full`, 495 and 496 on `new_ex0dte`; exact McNemar test $p = 1.000$ in both). On the published inputs they agree in 99.2% of the 1,476 windows (kappa 0.984), with 6 of the 12 differences resolved in favour of each.

With relative labels gpt-oss-20b detects 185 of the 223 windows of the 2024 baseline (83.0%) and with the dates 183 (82.1%); the two labelings differ in 6 windows, 4 detected with relative labels only and 2 with dates only (paired quarter-block bootstrap interval [0.0, 2.4] percentage points, $p = 0.633$; exact McNemar test $p = 0.688$ at stride one and $p = 1.000$ on the non-overlapping subsample of 8 windows), and in 2020 both labelings detect 27 of 223. With the dates it differs from the published o4-mini answers in 2 windows of 2024, both detected by gpt-oss-20b only, and in none of 2020. The 2024-versus-2020 contrast is 70.9 percentage points with relative labels ($\varphi = 0.709$) and 70.0 with dates ($\varphi = 0.701$), with the Fisher test on the non-overlapping subsample at $p = 0.041$ in both cases. The 2024 baseline with relative labels is the arm where gpt-oss-20b departs most from the rule (185 detections against 181; agreement 98.2%, kappa 0.939).

Repeated requests give the first verdict again in 74 of 74 bridge windows, 234 of 235 windows of the complete series, 241 of 241 windows without same-day expiries and 11 of 11 in each baseline arm. The reported metrics match the computed ones about as well as those of gpt-6-luna: on the complete series the reported mean |GEX| lies within 5% of the computed value in 99.5% of the answers (gpt-6-luna 98.7%), and the sign-flip count is exact in 93.4% (gpt-6-luna 93.4%).

Table S6: gpt-oss-20b: detection rate in percent (detected windows in brackets) of the deterministic rule, of gpt-6-luna and of gpt-oss-20b on the same windows, and the agreement of gpt-oss-20b with gpt-6-luna and with the rule in percent (Cohen's kappa in brackets). Windows are those with a parsed answer of gpt-oss-20b; the rule and gpt-6-luna rates are over all windows of the arm.

Arm	Windows	Rule	gpt-6-luna	gpt-oss-20b	vs gpt-6-luna	vs rule
`oss_bridge_jan`	1,476	45.2 (667)	45.5 (671)	45.6 (673)	99.2 (0.984)	99.6 (0.992)
`oss_full_spy`	4,689	13.0 (609)	13.2 (617)	13.2 (617)	99.4 (0.973)	99.7 (0.988)
`oss_ex0dte_spy`	4,691	10.5 (491)	10.6 (495)	10.6 (496)	99.6 (0.976)	99.9 (0.992)
`oss_p34_obf_2024`	223	81.2 (181)	80.7 (180)	83.0 (185)	97.8 (0.925)	98.2 (0.939)
`oss_p34_dated_2024`	223	81.2 (181)	81.2 (181)	82.1 (183)	99.1 (0.970)	99.1 (0.970)
`oss_p34_obf_2020`	223	12.1 (27)	12.1 (27)	12.1 (27)	100.0 (1.000)	100.0 (1.000)
`oss_p34_dated_2020`	223	12.1 (27)	12.1 (27)	12.1 (27)	100.0 (1.000)	100.0 (1.000)

S3.2 Llama 3.1 8B

Every answer of Llama 3.1 8B parses at the first attempt and none comes near the token limit (about 170 completion tokens on average). The one window that it does not detect in each complete series is the same window, ending 2010-01-28. Because it detects nearly every window,

its agreement with the rule equals the detection rate of the rule (45.2% on the bridge, 13.0% and 10.5% on the complete series, 81.2% and 12.1% on the baselines), with kappa 0.000 in every arm, and repeated requests return the first verdict in all 588 repeated windows. Its reported mean |GEX| lies within 5% of the computed value in 14.3% to 64.1% of the answers per arm and its sign-flip count is exact in 2.7% to 13.9% (Table S7). For the window of `new_full` ending 2008-02-27 it reports 10 positive and 20 negative days and writes that "the persistence percentage of 66.67% meets the 70% threshold" and that "the average magnitude of $2.45B is greater than the $5B threshold".

Table S7: Llama 3.1 8B: windows, windows detected, agreement with the rule in percent (Cohen's kappa in brackets), the share of answers whose reported mean |GEX| lies within 5% of the computed value, the share whose sign-flip count is exact, and the repeated requests that returned the first verdict. Every answer parsed.

Arm	Windows	Detected	vs rule	Mean |GEX| within 5%	Flips exact	Repeats equal
`llama_bridge_jan`	1,476	1,476	45.2 (0.000)	41.6	12.1	74 of 74
`llama_full_spy`	4,691	4,690	13.0 (0.000)	19.1	11.0	235 of 235
`llama_ex0dte_spy`	4,691	4,690	10.5 (0.000)	16.1	13.1	235 of 235
`llama_p34_obf_2024`	223	223	81.2 (0.000)	64.1	6.3	11 of 11
`llama_p34_dated_2024`	223	223	81.2 (0.000)	62.3	5.8	11 of 11
`llama_p34_obf_2020`	223	223	12.1 (0.000)	14.3	13.9	11 of 11
`llama_p34_dated_2020`	223	223	12.1 (0.000)	23.8	2.7	11 of 11

S3.3 Qwen3-8B

Table S8 gives the share of windows that Qwen3-8B answered in each arm, split by the verdict of the rule, its agreement on the answered windows and the bounds of its detection rate. On the 2020 baselines it answers 223 and 218 of the 223 windows; on the 2024 baseline it answers 33 of the 42 windows that the rule rejects with relative labels and 19 of them with dates. On the published inputs it answers 1,287 of the 1,412 published windows and differs from o4-mini in 14 (12 detected by Qwen3-8B only), and in all 14 the rule sides with o4-mini (exact McNemar test at stride one $p = 0.013$). With the dates it differs from the published o4-mini answers in 9 of the 197 windows of 2024 that it answered, all detected by Qwen3-8B only ($p = 0.004$), and in 1 of the 218 of 2020.

Of the 99 windows that Qwen3-8B detects and the rule rejects, 83 fail only the limit of five sign flips: they have six to ten flips (52 have six), and Qwen3-8B reports three to five. Of the others, 14 fail only the magnitude criterion (9 of them with a computed mean |GEX| between $4.96 and $5.00 billion, which Qwen3-8B reports as $4.97 to $5.00 billion and accepts), 1 fails only the persistence criterion and 1 fails both the magnitude criterion and the flip limit. In the 9 windows that the rule detects and Qwen3-8B does not, it reports a mean |GEX| below $5 billion where the computed mean is above it in 7, a lower persistence in 1 and more sign flips in 1. Its reported mean |GEX| lies within 5% of the computed value in 96.9% to 99.5% of the answered windows per arm, and its sign-flip count is exact in 86.3% to 100.0% (87.3% on the complete series, against 93.4% for gpt-6-luna). Repeated requests give the first verdict again in 57 of 57 bridge windows, 193 of 195 windows of the complete series, 196 of 197 without same-day expiries and 34 of 34 in the baselines, counting the windows answered in both passes.

Table S8: Qwen3-8B: windows of each arm, the share answered within 8,000 completion tokens (all windows, and the windows that the rule detects and rejects), the agreement with the rule and with gpt-6-luna on the answered windows (Cohen's kappa in brackets), and the bounds of the detection rate over all windows (lower: every unanswered window counted as not detected; upper: as detected), next to the rate of the rule. All values in percent.

		Answered			Agreement		Detection rate	
Arm	Windows	All	Rule detects	Rule rejects	Rule	gpt-6-luna	Bounds	Rule
`qwen_bridge_jan`	1,476	91.1	99.9	83.9	98.9 (0.978)	98.6 (0.972)	45.9–54.7	45.2
`qwen_full_spy`	4,691	89.7	98.9	88.3	98.7 (0.948)	98.5 (0.941)	13.8–24.2	13.0
`qwen_ex0dte_spy`	4,691	92.0	98.6	91.3	99.4 (0.973)	99.3 (0.966)	10.7–18.7	10.5
`qwen_p34_obf_2024`	223	95.1	98.9	78.6	98.1 (0.924)	97.6 (0.907)	82.1–87.0	81.2
`qwen_p34_dated_2024`	223	88.3	98.3	45.2	95.4 (0.668)	95.4 (0.668)	83.9–95.5	81.2
`qwen_p34_obf_2020`	223	100.0	100.0	100.0	100.0 (1.000)	100.0 (1.000)	12.1	12.1
`qwen_p34_dated_2020`	223	97.8	96.3	98.0	99.5 (0.979)	99.5 (0.979)	12.1–14.3	12.1